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EXERCISE SHEET / THEME 3: QUANTITATIVE VARIABLES.

EXERCISE 1. COMMUTE TIME FROM HOME TO THE UNIVERSITY.

You survey 15 students of your class and ask them how long it takes to get from their house to the university. Their answers are expressed in minutes and are the following:

3	12	17	20	22	22	28	33	34	35	35	37	42	46	58

a) What is the range, the median and the mean of the distribution?
b) Create a frequency table using ranges of 10 units each, starting at 0 (i.e. 0–9, 10–19, 20-29, and so on). The table should include the following columns:
   i. Range,
   ii. frequency,
   iii. percentage and
   iv. cumulative percentage.
c) With the information presented in the frequency table that you created, build a histogram with the same ranges.
d) How many students take less than half an hour to get to the university?

EXERCISE 2. HOURLY WAGE.

Now you survey 15 applicants to a social transfer and ask them about their hourly wage. The answers are expressed in EUR and are the following:

4	3	5	6	14	18	22	28	33	35	48	51	53	81	85

a) What is the range, the median and the mean of the distribution?
b) Build a histogram using ranges of 20 units each, starting at 0 (i.e. 0–19, 20–39, and so on).
c) How many earn at least 60 EUR per hour?
d) How many earn between 20 and 59 EUR per hour?

EXERCISE 3. COMPARISON BETWEEN DISTRIBUTIONS.

Considering the information from both histograms (exercises 1 and 2).

a) What can you say about the symmetry of each distribution? The concept skewness can be useful (is any of them right or left skewed?).
b) What can you say about the mean and median as measures used to describe each distribution? Is any of them more limited than the other in some cases?